Electric field due a point charge

⇒ **Field** (Temperature has definite value at room) ↘ Thermometer
(Pressure of fluid in pipe)

⇒ can represents mathematically by T(x, y, z)
or by graph. (of T & P) (Do not vary with time then static field)

g (gravitational field) $g = \frac{F}{m_0}$ (vector field)

⇒ mass ⇌ (interact) mass (Before concept of field interact directly and instantaneously)

⇒ mass ⇌ field ⇌ mass (graviton)

⇒ (mass interact not directly with other instead with gravitational field established by the other) (mass field certain value at every point)

⇒ field plays a role of intermediary.

Electric field Force exerted on test charge q_0.

$$\vec{E} = \frac{\vec{F}}{q_0} \quad (N/C)$$

q_0 = test charge

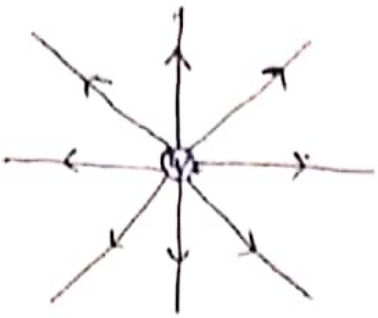
⇒ The direction of $\vec{E}$ is same as direction of F

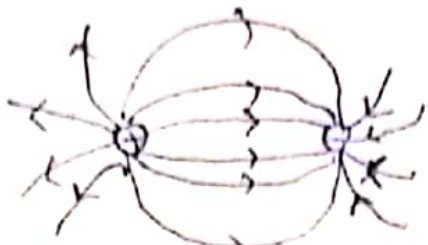
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Electric field Lines.

=> Introduced by Michael Faraday himself

Electric field





Properties of Electric field lines

(i) Originate from positive(+ve) and terminate at negative charge(-ve)

=> electric field lines are continous and smoth

=> Two electric field lines not cross each other or intersect each other

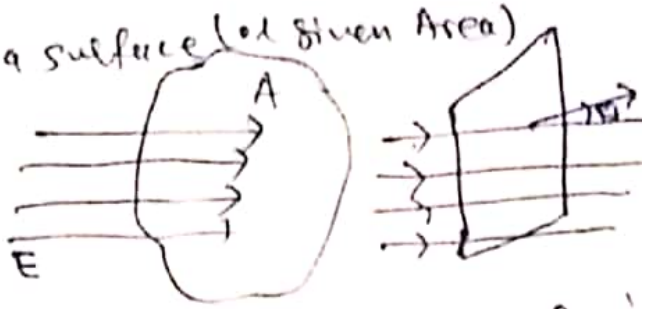
=> when field is strongest then field lines are close to eachother

Electric Flux

"Electric field lines passing through a surface (of given Area)

$$\phi_E = E \cdot dA$$

$$\phi_E = EA\cos\theta$$

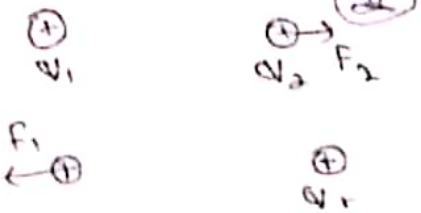


If $\theta = 90°$, That is E and A are ⊥, $\phi_E = 0$ (min

$\theta = 0°$, // // // E // A are parallel ϕ_E = max

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Two force would be equal ($F_1 = -F_2$) even though the two electric field may be quite diffelent.



Electric field due to a point charge

⇒ Let positive test charge q_0 a distance r away from point charge

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

If we have N point charge

$$E = E_1 + E_2 + E_3 + \cdots$$

Electric Dipole

Two charges with opposite ~~charges~~ with equal magnitude and seperated by a distance d"

$$P = qd$$

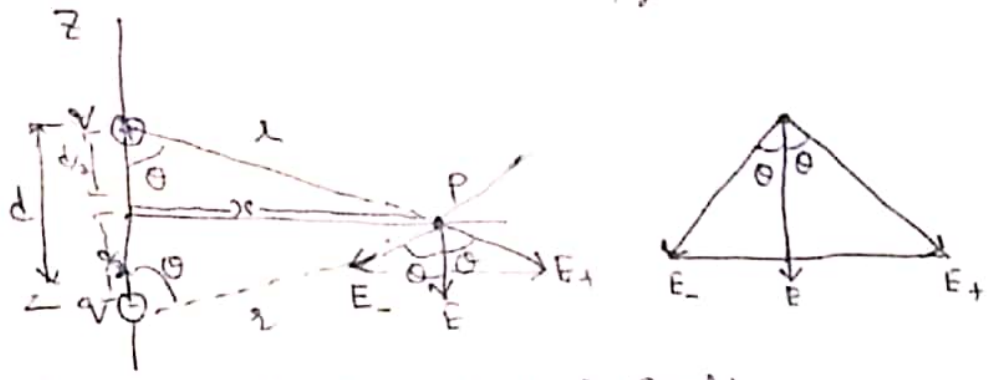
P = electric dipole moment

$$E = E_+ + E_- \quad \text{——} (1)$$

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$$E_+ = E_- = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{q}{x^2+(d/2)^2} \quad -(2)$$

(3)



x-component of the total field

$$E_+ \sin\theta - E_- \sin\theta = 0$$

The total electric field only a z-component

$$E = E_+\cos\theta + E_-\cos\theta = 2E_+\cos\theta \quad -(3)$$

$$\cos\theta = \frac{d/2}{\sqrt{x^2+(d/2)^2}} \quad -(4) \quad \left(\because r^2 = x^2+\left(\frac{d}{2}\right)^2\right)$$

using eq. (2) and (4) in eq. (3)

$$E = 2\,\frac{1}{4\pi\epsilon_0}\,\frac{q}{x^2+(d/2)^2}\cdot\frac{d/2}{\sqrt{x^2+(d/2)^2}}$$

$$E = \frac{1}{4\pi\epsilon_0}\cdot\frac{qd}{\left[x^2+(d/2)^2\right]^{3/2}}$$

$\Rightarrow$ Field is directly proportional to qd

$(\because P = qd)$

$$E = \frac{1}{4\pi\epsilon_0}\cdot\frac{p}{x^3}\,\frac{1}{\left[1+\left(\frac{d}{2x}\right)^2\right]^{3/2}}$$

using Binomial Expansion

$$(1+y)^n = 1+ny+\frac{n(n-1)}{2!}y^2+\cdots\cdots$$

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$$E = \frac{1}{4\pi\epsilon_0}\frac{p}{x^3}\left[1+\left(\frac{d}{2x}\right)^2\right]^{-3/2} \quad (4)$$

$$E = \frac{1}{4\pi\epsilon_0}\frac{p}{x^3}\left[1+(-3/2)\left(\frac{d}{2x}\right)^2+\cdots\right]$$

$$E = \frac{1}{4\pi\epsilon_0}\frac{p}{x^3}$$

=> Dipole moment is the fundamental property of the molecule (which contain +ve -ve charge)

☆ Micheal Faraday gives the idea of **lines of force**

=> lines of forced give direction E

=> " " " originate from +ve and terminate on -ve charge

=> No. of lines per unit area is ∝ to magnitude of E field.

The electric charge of continous charge distributions.

=> Even though electric charge is quantized a collection of large numbers of elementary charges can be regarded as continous charge distribution.

=> E can computed in continous charge distribution by dividing it infinitesimal elements dv.

=> dv establish field dE at point P. $E = \int dE$ Resultant field at P by

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$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$ (5)

⇒ A continous charge distribution is described by charge density.

$\lambda = \frac{dq}{ds}$ ⇒ $\boxed{dq = \lambda\, ds}$ (for linear charge distribution)

λ = linear charge density or (charge per unit length)

⇒ If object is uniformly charged then

$dq = \frac{Q}{L} ds$ (Q = Total charge)

$\sigma = \frac{dq}{da}$ $\boxed{dq = \sigma\, da}$

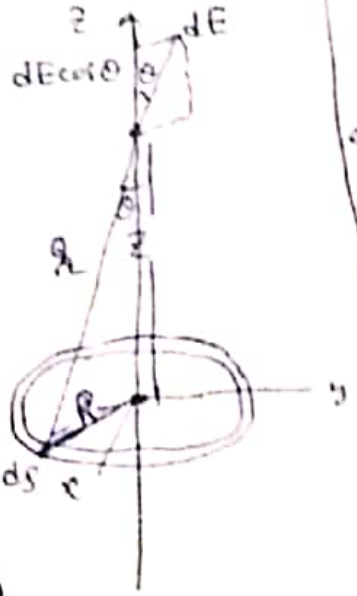
σ = surface charge density (or charge per unit area)

$\rho = \frac{dq}{dv}$ $\boxed{dq = \rho\, dv}$ (Volume charge density)

☞ **Ring of charge** $dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{s^2} = \frac{\lambda ds}{4\pi\epsilon_0 (z^2+R^2)}$

⇒ consider Ring of Radius R having linear charge density λ around its circumference

⇒ consider ring to be made of plastic (or) some other insulator (so charged can be regarded as fixed at place)



⇒ consider two elements of charge on ring located at opposite end of diameter

⇒

⇒ Net $\vec{E}$ is pallel to z-axis as components ⊥ to axis cancel each other.

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$$\cos\theta = \frac{z}{r} = \frac{z}{(z^2+R^2)^{1/2}} \quad \text{—(2)}$$

(6)

$$dE_z = dE\cos\theta = \frac{z\lambda\, ds}{4\pi\epsilon_0 (z^2+R^2)^{3/2}} \quad \text{—(3)}$$

$$E_z = \int dE\cos\theta = \frac{z\lambda}{4\pi\epsilon_0 (z^2+R^2)^{3/2}} \int ds$$

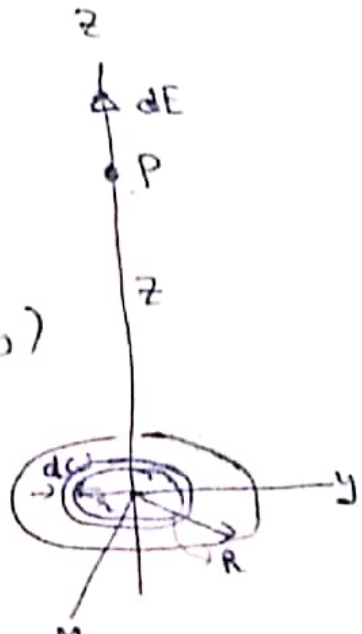
$$E_z = \frac{z\lambda\,(2\pi R)}{4\pi\epsilon_0 (z^2+R^2)^{3/2}} \quad \text{—(4)}$$

$$E_z = \frac{qz}{4\pi\epsilon_0 (z^2+R^2)^{3/2}} \quad \text{—(5)} \qquad \left(\begin{matrix} q = \lambda l \\ q = \lambda(2\pi R)\end{matrix}\right)$$

If $z >> R$, then eq. (5) becomes:

$$E_z = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \qquad (z >> R)$$

A Disk of charge



$dq = \sigma dA = \sigma(2\pi\omega)d\omega$

$dA = 2\pi\omega d\omega$ is differential area of the ring.

$$dE_z = \frac{z\sigma(2\pi\omega)d\omega}{4\pi\epsilon_0 (z^2+\omega^2)^{3/2}} \quad \text{(similarly)}$$

$$dE_z = \frac{\sigma z}{4\pi\epsilon_0} (z^2+\omega^2)^{-3/2}(2\omega)d\omega$$

$$E_z = \int dE_z = \frac{\sigma z}{4\pi\epsilon_0} \int_0^R (z^2+\omega^2)^{-3/2}(2\omega)\, d\omega$$

$$X = z^2+\omega^2, \quad m = -3/2, \quad dX = (2\omega)d\omega$$

$$E_z = \int X^m dX$$

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$$\int x^n dx = \frac{x^{n+1}}{n+1} = \frac{x^{-3/2+1}}{-3/2+1} = \frac{x^{-1/2}}{-1/2}$$

$$E_z = \left| -\frac{2}{x^{1/2}} \right|_0^R \times \frac{\sigma z}{4\pi\epsilon_0}$$

$$E_z = \left(\left| -2 (z^2+\omega^2)^{-1/2} \right|_0^R \right) \frac{\sigma z}{4\pi\epsilon_0}$$

$$E_z = \frac{\sigma z}{4\pi\epsilon_0} \left[-2(z^2+R^2)^{-1/2} + 2\left(\frac{1}{z}\right) \right]$$

$$= \frac{\sigma z}{2\epsilon_0} \left[\frac{1 - z(z^2+R^2)^{-1/2}}{z} \right]$$

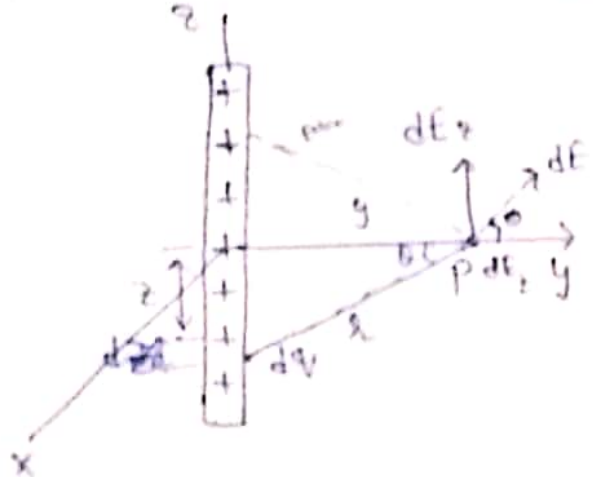
$$E_z = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{(z^2+R^2)^{1/2}} \right)$$

(charged disk)

If $z > 0$ and, $R \gg z$

$$E_z = \frac{\sigma}{2\epsilon_0}$$

Infinite line charge



$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\lambda\, dz}{y^2+z^2}$$

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$dE_y = dE\cos\theta$ $\quad dE_z = dE\sin\theta$ (8)

$$E_y = \int dE_y = \int_{z=-\infty}^{z=+\infty} \cos\theta\, dE$$

$$E_z = \int dE_z = \int_{z=-\infty}^{z=+\infty} \sin\theta\, dE$$

E is entirely along y-axis.

⇒ strictly if y-axis passes through the middle of line

⇒ E due to upper and lower half

$$E = E_y = 2\int_{z=0}^{z=\infty} \cos\theta\, dE$$

$$E = \frac{\lambda}{2\pi\epsilon_0} \int_{z=0}^{z=\infty} \cos\theta \frac{dz}{y^2+z^2}$$

$$z = y\tan\theta$$

$$dz = y\sec^2\theta\, d\theta$$

$$E = \frac{\lambda}{2\pi\epsilon_0} \int_{\theta=0}^{\theta=\pi/2} \cos\theta \frac{(y\sec^2\theta)\, d\theta}{y^2(1+\tan^2\theta)}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 y} \int \cos\theta\, d\theta$$

$$E = \frac{\lambda}{2\pi\epsilon_0 y} \left|\sin\theta\right|_0^{\pi/2}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 y}$$

$$\boxed{E = \frac{\lambda}{2\pi\epsilon_0 x}}$$

(infinite line)

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